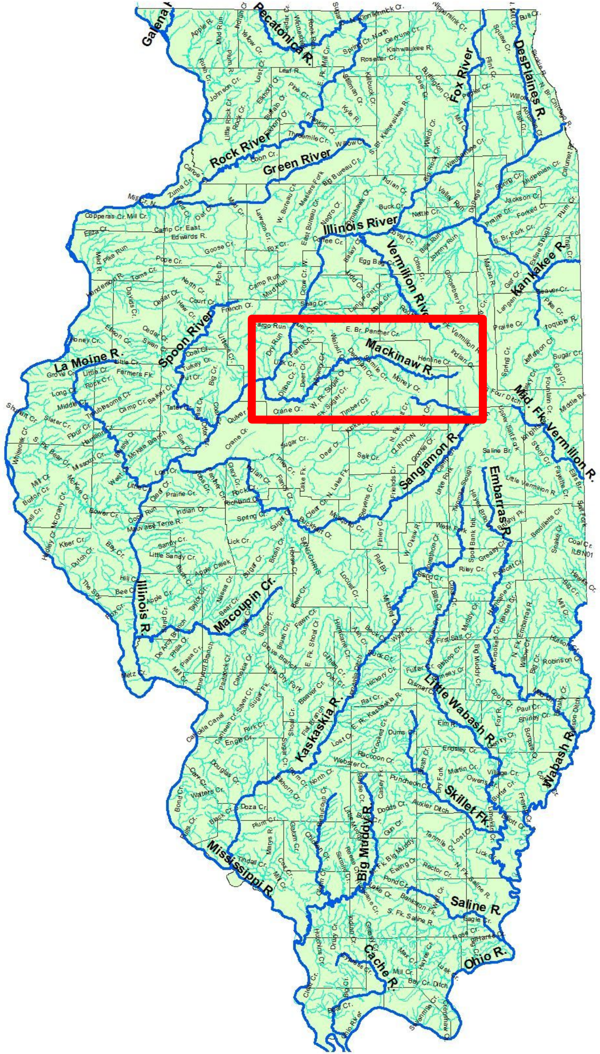
Mackinaw River, IL, Stressor Analysis

By Caitlyn Feldpausch
EAS-519 Final



Area of Concern (AOC) and General Info

- 3.000 km^2 watershed, major tributary of the Illinois River which drains into the Mississippi River within McLean, Woodford, and Tazewell counties.
- 60-70 native fish species and 25-30 mussel species.
- It was selected as a priority site by the Nature Conservancy (TNC) more than 25 years ago due to it being a high-quality river sustaining some of the highest-quality streams in Illinois.
- Top landcover types include: cultivated crops, deciduous forest, hay/pasture, low intensity development, developed open space, and medium intensity development.
- Watershed mainly used for agriculture

Mackinaw River Watershed Map:

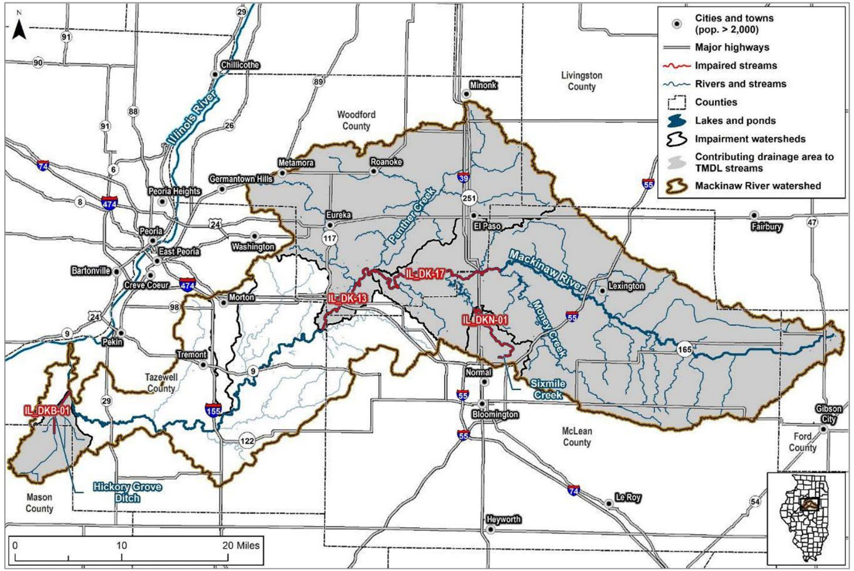
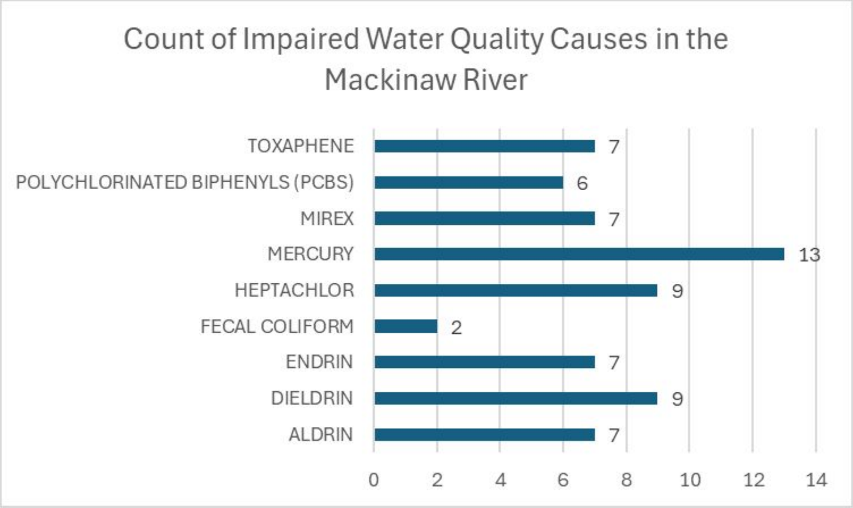


Figure 1. Mackinaw River watershed, TMDL project area.
Note: IL_DKB-01 and IL_DKN-01 are not addressed in this TMDL document. See Appendix C for more information.

303(d) Listed Chemical Stressors

Potential Source	Relative Fecal Coliform Contributions to Mackinaw River (IL_DK-13)
Agricultural runoff	High (animal agriculture)
Stormwater runoff	Low
Onsite wastewater treatment systems	Low
Wildlife	Moderate

Agricultural Runoff is the highest cause of fecal coliform levels in the Mackinaw River by the 303(d) report.



The number of times water quality issues were listed in the most recent Illinois Integrated Water Quality Report 303(d) list (2024).

303(d) Listed Chemical Stressor Information

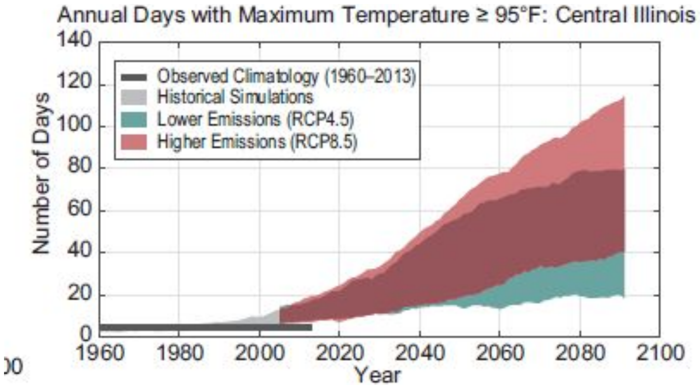
- Mercury, Heptachlor, and Dieldrin, are the top 3 causes of water quality impairment from the TMDL Total Maximum Daily Load report.
- Mercury is a rare earth element still used in some quantities despite known toxicity, EPA standards. Heptachlor (banned in ‘78), Mirex, and Dieldrin (banned in ‘78) were used as insecticides. Mirex was also used as a fire retardant for plastic and rubber.
- Due to their non-flammability, chemical stability, high boiling point and electrical insulating properties, man-made organic PCB chemicals were used in hundreds of industrial applications like in plastic and rubber products, banned in 1979.
- Endrin, Aldrin, and toxaphene were used as insecticides or pesticides.
- There are 16 species of total coliform are in soils, plants, and animal and human waste. Fecal coliform bacteria is a subgroup that grows at higher temperatures and are found only in warm-blooded animal fecal waste. One of its 6 species is E. coli. Agricultural runoff has the greatest impact on fecal coliform levels in the river according to a recent Total Maximum Daily Load report table shown.

303(d) Listed Stressor Societal Impacts

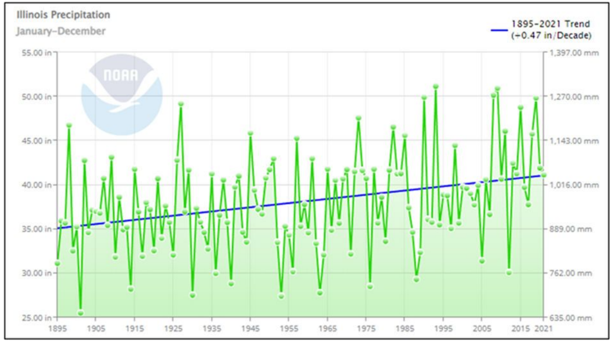
- Mercury is one of the world’s most toxic metals and is a neurotoxin. Mercury levels have caused a statewide fish consumption advisory due to predator fish concentrations. Mercury can harm wildlife that eats contaminated fish, particularly bald eagles, loons and other fish-eating birds and mammals. This is based on recent studies showing adverse effects to the developing nervous system of fetuses possibly resulting in low IQ, abnormal muscle tone, slowed motor function. Fish affected include black bass like largemouth bass, walleye, catfish, northern pike, and more. The advisory is meant for sensitive populations like pregnant women and kids under 15.
- Heptachlor, banned in 1978, has been linked to neurological, behavioral, and immune problems in wildlife. Yet, it’s still allowed to control fire ants near transformers.
- Dieldrin has been linked to neurological, behavioral and immune-suppression problems in wildlife.



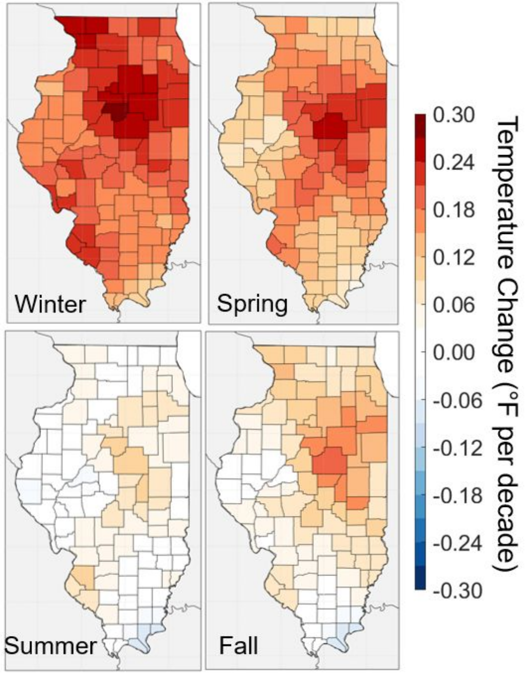
Climate Change Stressors & Outlook



Central IL will have more days above 90 degrees F, which started increasing more rapidly since about 2000.



Total annual IL precipitation has increased by nearly 6 inches between 1895 and 2019, an approximate 15% increase, and is expected to continue increasing with more intense rainfall events projected.



Observed daily average temperatures between 1990 and 2020 had become increasingly hotten, especially through winter, increasing the harvest season.

Human-Induced Stressors Not Fully in Report

- Agricultural runoff was already stated as a leading cause of bacteria loading, but also for nutrient and pollutant loading from farming practices.
- Tile-drainage systems have negatively impacted water quality and increased habitat loss.
- Tile-drainage systems involve a network of pipes connected to a main pipe for field drainage. It changes the field hydrology, carrying away as much phosphorus as surface runoff into waterways from animal-based manures. As climate change increases precipitation, times of saturated soils in undrained fields increase which decrease crop yields. So, there becomes a greater need for field tile drainage which increases phosphorus loss into streams. This then feeds and grows algal blooms which like phosphorus rich freshwater.
- Urban development causes higher pollution from septic drainage systems and urban storm runoff.

Increased times of highly saturated soils in undrained fields from increased precipitation → Greater need for tile drainage → Increased phosphorus runoff into waterways from farms → Increased algal blooms fed by phosphorus

Stressor Points Ranking Process

The following slide equations are how I implemented a points ranking system to rank stressors' cumulative impacts on water quality, and thus the local ecosystem.

Although chemicals likely do not directly increase concentrations of each other, conditions can cause additive, synergistic, or antagonistic stressor interactions, and thus increased or decreased concentrations of one chemical pollutant. For example, PCBs do not directly increase Dieldrin in water, but that does not mean some stressor interaction is taking place to affect both chemicals. This lets us assume that all stressor interactions: additive, synergistic, or antagonistic, are possible and lets us simplify equations. Our equations for calculating each stressor interaction also let us remove one of the interactions later on if an analyst knows it is not present, and they can recalculate cumulative stressor impacts.

The rank value can be doubled later if the listed 303(d) contaminant is also affected by one of the human-induced stressors listed and/or climate change stressors. Phosphorus will be added to the final ranking as a leading cause from manure or urban runoff as a primary human-induced stressor for the river.

Magnitude, Frequency, and Duration Terms Going Into Each Stressor Equation from 303(d) Report Tables

Total stressor interaction Impact =

Additive Stressor Interaction Cumulative Impact **(ACI) +**
Synergistic Stressor Interaction Cumulative Impact **(SCI) +**
(- Antagonistic Stressor Interaction Cumulative Impact **(ANCI))**

Frequency: Number of times “Cause” column entries occur (such as the number of times Toxaphene occurs, which is 7).

Duration: Use the pollutant highest half life in years, if any. This also provides a way of understanding whether pollutants will leave the system faster such as with pulse events, or if they will take years to leave such as with press events.

Magnitude: We do not have an amount of toxins in the water, but we have water body sizes where toxins were collected in data reports. Will be values based on “Water Size” column value as (smaller bodies cannot absorb temperature changes, acidity changes, less water volume to dilute toxins/pollution, affected by human factors such as industrialization more quickly).

Stressor Points Ranking Process/Equations

Total stressor interaction Impact =

Additive Stressor Interaction Cumulative Impact **(ACI) +**
Synergistic Stressor Interaction Cumulative Impact **(SCI) +**
(- Antagonistic Stressor Interaction Cumulative Impact **(ANCI))**

ACI = = **(**magnitude_s1 + frequency_s1 + duration_s1**) + (**magnitude_s2 + frequency_s2 + duration_s2**)**

SCI = ACI **+ (**(magnitude_s1 x magnitude_s2) + (frequency_s1 x frequency_s2) + (duration_s1 x duration_s2)**)**

ANCI = ACI **- (**(magnitude_s1 x magnitude_s2) + (frequency_s1 x frequency_s2) + (duration_s1 x duration_s2)**)**

…and so on between stressors 1,2,3,...etc.

Again, rank value can be doubled later if listed 303(d) contaminant is also affected by one of the human-induced stressors listed and/or climate change stressors. Phosphorus will be added to the final ranking as a leading cause coming from manure or urban runoff as a primary human-induced stressor for the river.

Link to calculations doc:
https://docs.google.com/document/d/11M7LqxLolebX0TP8lTOA_iDGlkmEbCBeC0x5iie0cVs/edit?usp=sharing

Example Calculation: PCBs on PCBs as Stressors 1 and 2

Priority	Hydrologic Unit Code	Water Body Name	Assessment Unit ID	Water Size*	Designated Use	Cause
Medium	713000407	Mackinaw River	IL_DK-04	10.42	Fish Consumption	POLYCHLORINATED BIPHENYLS (PCBS)
Medium	713000408	Mackinaw River	IL_DK-12	29.53	Fish Consumption	POLYCHLORINATED BIPHENYLS (PCBS)
High	713000303	Mackinaw River	IL_DK-12	29.53	Fish Consumption	POLYCHLORINATED BIPHENYLS (PCBS)
Medium	713000401	Mackinaw River	IL_DK-20	21.68	Fish Consumption	POLYCHLORINATED BIPHENYLS (PCBS)
Medium	713000403	Mackinaw River	IL_DK-20	21.68	Fish Consumption	POLYCHLORINATED BIPHENYLS (PCBS)
Medium	713000401	Mackinaw River	IL_DK-21	23.21	Fish Consumption	POLYCHLORINATED BIPHENYLS (PCBS)

PCBs: Magnitude = (**10.42 + 29.53 + 21.68 + 23.21) = 84.84** Frequency = **6 times** Duration = at most 15 years by quich NIH/Sciencedirect literature search

So,

ACI = (84.84 + 6 + 15) + (84.84 + 6 + 15) = 211.68
SCI = 211.68 + (84.84 x 84.84 + 6 x 6 + 15 x 15) = 7670.506
ANCI = 211.68 - (84.84 x 84.84 + 6 x 6 + 15 x 15) = -7247.146

And total additive, synergistic, and antagonistic stressor interaction impact = 211.68 + 7670.506 + -7247.146 **= 635.04**

Final Stressor-On-Stressor Impact Matrix

The next slide is the final matrix of individual stressor impact calculations, including additive, antagonistic, or synergistic effects. An interesting find was that some of these were not expected to be able to migrate to water when I was looking up half lives.

We sum rows or columns to get total cumulative impact values. Aldrin has a much greater stressor impact from aditivite, antagonistic, and synergistic interactions being possible between itself and all other chemicals. Mercury and PCBs follow with greater impact also. Fecal coliform has the lowest priority stressor point ranking due to an existing management plan which will be discussed later. The rest of the chemicals have limited variability between ranked points and thus fall farther down our priority list of what to mitigate first.

These highly impactful pollutant stressors match our previous qualitative societal impact information as PCBs are probable carcinogens and Mercury can cause poisoning, neural problems, and even death. In one incident, aldrin-treated rice killed hundreds of shorebirds and waterfowl along the Texas Gulf Coast and the fatal dose for an adult male is about 5 grams. We will now take into account human and climate stressors in the system. In the final ranking, all pollutants with low point variability in the 6000s and a point gap below the top two or three in this initial ranking, will move down the list in management priority.

303(d) Listed Stressor On Stressor Cumulative (ACI + SCI - ANCI) Impact

	Toxaphene	PCBs	Mirex	Mercury	Heptachlor	Fecal Coliform	Endrin	Dieldrin	Aldrin
Toxaphene	395.04	515.04	382.95	625.96	424.32	237.94	394.95	438.99	3180.69
PCBs	515.04	635.04	502.95	745.96	544.32	357.94	514.95	558.99	3300.69
Mirex	382.95	502.95	370.86	613.87	412.23	225.85	382.86	426.90	3168.60
Mercury	625.96	745.96	613.87	856.88	655.24	468.86	625.87	669.912	3411.61
Heptachlor	424.32	544.32	412.23	655.24	453.60	267.22	424.23	468.27	3209.97
Fecal Coliform	237.94	357.94	225.85	468.86	267.22	80.84	237.85	281.89	3023.59
Endrin	394.95	514.95	382.86	625.87	424.23	237.85	394.86	438.90	3180.60
Dieldrin	438.99	558.99	426.90	669.912	468.27	281.89	438.90	482.94	3224.64
Aldrin	3180.69	3300.69	3168.60	3411.61	3209.97	3023.59	3180.60	3224.64	5966.34

Sum Rows or Columns per Stressor for Final Stressor Cumulative Impact

Greatest to least total cumulative impact between each stressor, itself, and all other 303(d) listed stressors:

Aldrin = 31,666.7

Mercury = 8,674.16

PCBs = 7,675.88

Dieldrin = 6,991.43

Heptachlor = 6,859.4

Toxaphene = 6,595.88

Endrin = 6,595.07

Mirex = 6,487.07

Fecal Coliform = 5,181.98

To Note: Climate Change Stressors On Human-Induced Stressors

Increased storm frequency and precipitation amounts will naturally increase urban and agricultural runoff from storms. Increasing time of flooded fields will need to be drained more often by already used tile-drainage methods.

Increased urban development and urban expansion into rural or freshwater areas will further add chemicals like phosphorus into the river from increased urban runoff, sewage plants, and septic systems.

Human-Induced Added Stressors On 303(d) Listed Stressor Interactions

<table>
<tr><th></th><th>Toxaphene
PCBs
Mirex
Heptachlor
Endrin
Dieldrin</th><th>Mercury</th><th>Fecal Coliform</th></tr>
<tr><td>Agriculture Runoff</td><td rowspan="3">Have been banned.
*Agricultural runoff can still cause these banned chemicals to transport from soils to the water, to what degree could not be fully determined.</td><td></td><td rowspan="2">Is present in agricultural runoff, further induced by climate change precipitation effects.</td></tr>
<tr><td>Tile-Drainage</td><td></td></tr>
<tr><td>Urban Development</td><td>Can increase from Fish consumption enhanced by urban development</td><td></td></tr>
</table>

Human caused stressors for the area cannot really affect Toxaphene, PCBs, Mirex, Heptachlor, Endrin, or Dieldrin. However, agricultural runoff can still cause these banned chemicals to transport from soils to the water, but the extent of this could not be fully determined, mostly due to quantifying problems involving past use. Mercury effects can increase by urban development if people are not paying attention to fish consumption. and Fecal Coliform is present in agricultural runoff, further induced by increased frequency and magnitude of precipitation events.

Climate Change Stressors On 303(d) Listed Stressor Interactions

	Toxaphene	PCBs	Mirex	Endrin	Heptachlor	Dieldrin	Aldrin	Mercury	Fecal Coliform
Increased annual temps & warner winters.	All of these chemicals can evaporate as volatile organic compounds into the atmosphere from increasing temperatures. Volatile organic compounds are gases emitted into the air from products or processes. As these were harmful in the water, they are also harmful out of the water. Some are carcinogenic and some can react with other gases to form other air pollutants after evaporating. So, all of these pollutants would get doubled for rising temperature effects, but as they are all doubled, the ranking will stay the same as points change.							Fecal coliform grows more in higher temperatures which are being exacerbated by climate change. Mercury becomes toxic methylmercury in water. A 2018 study found methylmercury in zooplankton, eaten by many fish, grew 2-7x w/ increased temps.	

All of these chemicals would technically be doubled, either as volatile chemicals that can harm habitats and people as they evaporate and become airborne in increasing temperatures, or as they grow and become more toxic in water. Since we would double all chemicals ranked, even though the point values would change, our ranking stays the same.

Phosphorus

- Phosphorus increases with increased agriculture or urban runoff from climate change induced, higher frequency and magnitude flood and precipitation events.
- It is widely used in detergents, industrial processes, fertilizers, and is widely found in compounds and minerals.
- White phosphorus is very toxic, it can make you sick and can harm eyes or the respiratory system as a smoke.
- It can speed up eutrophication in water and does not dissolve well in water.

Adjusted Cumulative Impact Values per Pollutant Stressor After Determining Any Climate/Human Stressor Interactions Present

Stressors ranked from highest to lowest priority of mitigation based on ranking points and previous analysis:

1. **303(d) stressors and Phosphorus**

   A. Aldrin, B. Mercury, C. PCBs, D. Phosphorus,

   E. Dieldrin, F Heptachlor, G. Toxaphene, H. Endrin, I. Mirex, J. Fecal Coliform.

2. **Human induced stressors**

   A. Agricultural runoff, B. Urban development, C. Tile drainage systems.

3. **Climate stressors affecting human stressors and pollutants**

   A. Higher temperatures, B. More frequent precipitation events in greater magnitude.

Climate change stressors will be on the bottom of the list as they are very difficult and improbable to address fully at the local level. We can address its impacts on local level stressors such as the impact of greater precipitation on runoff adding pollutant concentrations. Human stressors move to the middle of the list as they can be mitigated through policies, although typically slowly, and are direct influences on further nutrient or pollutant runoff into the river. 303(d) stressors with phosphorus, and their rankings, thus stay at the top of the list for now. Phosphorus is added to the ranking below the highest ranked 303(d) stressors, but above the ones with low variability. Agricultural runoff, already listed as the top source for Fecal Coliform in the 303(d) report, stays as the main source as it will increase with more precipitation from climate change. Urban development is after that due to the river's proximity to small or moderate development. Higher temps are capable of adding all the pollutants or chemicals into the air through increased evaporation, with many being highly toxic in small amounts, so that climate stressor is first, as well as its ability to change water species densities. Then we have precipitation increasing urban/agricultural runoff.

Already Addressed Contaminants in a TMDL 2023 Report

- A Total Maximum Daily Load calculates "the maximum amount of a pollutant allowed to enter a waterbody so that the waterbody will meet and continue to meet water quality standards for that pollutant." A developed TMDL was published in 2023 for Nitrogen (Nitrate), and Fecal Coliform, to get those pollutants to water quality standards in the Mackinaw River. They were addressed due to a direct impact on recreation and public or food processing water supplies.

- **Developing a State of IL TMDL for a waterbody or watershed includes 3 stages:**
  1. Watershed characterization, historical data evaluation and analysis, picking a methodology, and data gap identification.
  2. Data collection to fill in gaps as necessary.
  3. Model calibration, TMDL scenarios, and an implementation plan.

TMDL Fecal Coliform Bacteria and Nitrogen (Nitrate) Report and Implementation Plan

- A developed 2023 TMDL report describes a 53% reduction in fecal coliform concentrations and a 44% reduction in nitrate loading necessary to meet water quality standards.

Needed management changes:

- Nutrient management.
- Filter strips, riparian buffers, bank stabilization and erosion control.
- Urban/residential nutrient reduction strategies.
- Private septic system inspection and maintenance program.
- Stormwater volume control.

2023 TMDL Fecal Coliform Bacteria and Nitrogen (Nitrate) Implementation Plan Needed Management Changes Further Explained

A reduction in bacteria and nutrient transport from fields and minimized soil loss is needed. Conservation tillage will control erosion & winter fertilizer use will be reduced, eliminating fertilizer near inlets, by installing stream buffers and filter strips.

Streambanks will be protected by planting vegetated or buffer areas with grasses, shrubs, or trees to mitigate bacteria inputs into surface waters. Priority will be given to areas with high erosion.

Reducing stormwater runoff from shoreline homes and increasing municipal programs like street sweeping will reduce runoff of nutrients and pollutants.

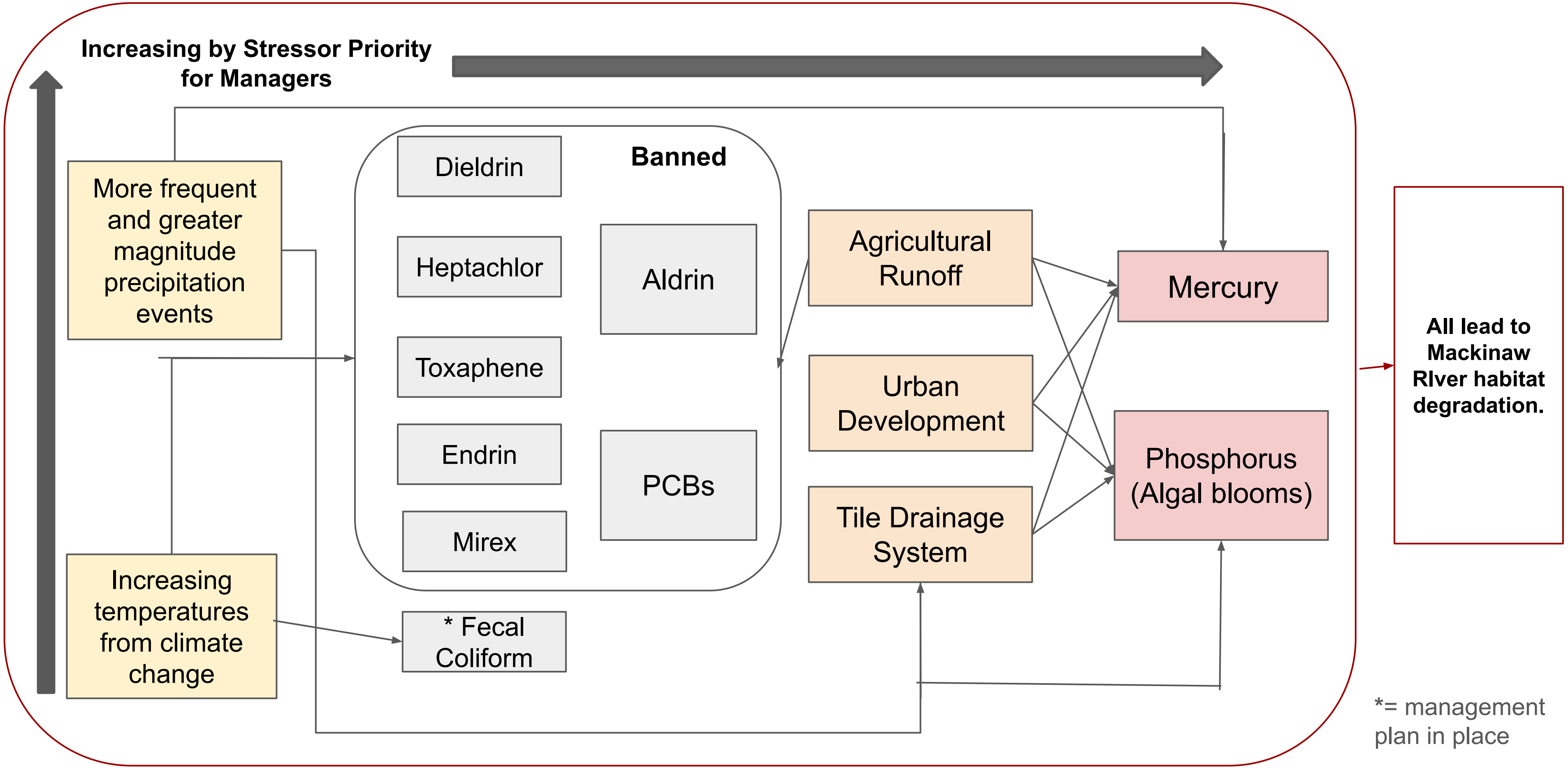
Failing septic systems and needed upgrades will be identified.

Rain gardens, bioretention areas, detention ponds, pervious pavement, and rain barrels will slow runoff. Green tech will let stormwater infiltrate, evaporate or evapotranspire before reaching the stormwater conveyance system

Final Ranking Explained

- Pollutant stressors are linked to human and/or climate stressors, Fecal coliform is given a star as it has an existing management plan with requirements known to meet water quality standards, and has a lowest stressor point ranking from the earlier calculation. Aldrin does not dissolve easily in water but has basically been banned, so Aldrin will move below Mercury, Phosphorus as the top two chemical stressors. PCBs will also move down as it has been banned. As a note from earlier, phosphorus will increase algal blooms that will create dead zones of no oxygen over time and clog fish gills. The banned pollutants will be presumed to go away with time, and Fecal Coliform is decreasing in priority already from its TMDL report implications. Mercury is in fluorescent lights, other products as regulated to certain standards, & has basically been banned, but due to the Illinois mercury water hazard warning and the amount of mercury in predator fish, it is still a top issue.
- Human stressors can more easily be dealt with then and should move up in priority, as they impact the top three 303(d) pollutants, along with climate change.
- Increasing precipitation is considered worse as it directly impacts larger magnitude human caused stressors already increasing existing 303(d) stressors.

Final Ranking Changed By Existing Plans (Visual Representation)



Further Restoration/Mitigation Methods and Uncertainty

- Recent official reports utilized recent water sample data, but old graphs from 2001, so Illinois EPA report graphs and figures, such as for weather factors, should be updated.
- An existing Nature Conservancy program with the Franklin Research and Demonstration Farm allows research on agricultural practices benefiting farmers and conservation practices. These practices should be further communicated with private property owners along the river.
- A Friends of the Chicago River Natural Solutions Tool incorporates healthy, equitable, biodiverse, protected, and connected landscapes and communities into its model with over 90 data layers. The tool lets you visualize and customize data for free to address complex environmental issues and substantiate work already being done. The public can make donations to help improve and sustain the tool online.
- Encourage community & private property use of nature-based solutions to help prevent runoff on private lands bordering the Mackinaw River: Natural Solutions Tool.
- Implement green materials in urban development.
- Possibly implementing green materials in urban development would be also help.
- Increase public meetings to address contaminants and increase awareness.

Economic/Political Effects on Stressors & Restoration

- EPA regulations and national regulations on chemicals involve long, drawn-out processes that struggle to induce action quickly in affected communities.
- The scope of EPA regulation ability on waterways has been challenged in court and recently substantially decreased for the Clean Water Act.
- Implementation of wetlands has seen water quality improvement, but could be costly depending on local budgets. In a 2016 analysis of developed wetland effects in the Mackinaw River watershed had the total cost of wetland installation from ~$15,000 to $49,000 for 1.0 to 4.5 acres, including a wetland pond & buffer. Yet, at one site installed, Arrowsmith North wetland, by 6 months the Total Suspended Solids reduced by 68%.
- There are grants for farmer sustainability training, and an IL grant for non-metropolitan communities wanting to improve public infrastructure alleviating detrimental health conditions, especially for water, sanitation, and storm sewer projects.

References

https://epa.illinois.gov/topics/water-quality/watershed-management/tmdls.html
https://epa.illinois.gov/topics/water-quality/watershed-management/tmdls/303d-list.html
MackinawRiverWatershed_Approval_DecDoc_TMDL_20231017.pdf
Mackinaw River Watershed | The Nature Conservancy in Illinois
Harnessing Nature to Filter Agriculture Runoff in Illinois | TNC
https://www.epa.gov/sites/default/files/2016-09/documents/heptachlor.pdf
https://greatlakesecho.org/2014/02/07/record-levels-of-banned-insecticide-found-in-illinois-otters/
https://www.scph.org/sites/default/files/editor/WQ/total%20coliform%20fact%20sheet.pdf
Learn about Polychlorinated Biphenyls | US EPA
https://semspub.epa.gov/work/05/238521.pdf
Endrin | Toxic Substances | Toxic Substance Portal | ATSDR
https://www.britannica.com/science/aldrin
Toxaphene | Organochlorine, Persistent Pollutant, Chlorinated Hydrocarbon | Britannica
https://epa.illinois.gov/topics/waste-management/materials-management/mercury/our-environment.html#:~:text=Mercury%20can%20harm%20wildlife%20that%20eats%20contaminated%20fish%2C,found%20in%20predator%20fish%20caught%20in%20Illinois%20waterways.
https://idph.illinois.gov/envhealth/fishadv/fishadvisory_qa.htm#:~:text=The%20Illinois%20Fish%20Contaminant%20Monitoring%20Program%20%28IFCMP%29%2A%20has,predator%20fish%20in%20Illinois%20waters%20due%20to%20methylmercury.
Climate Change in Illinois – Illinois State Climatologist

References

Changing Rainfall Patterns
Climate Explorer
Tile-drained Farm Fields Generally Increase Yields but Can They Protect Water Quality Too? | Lake Champlain Sea Grant
Tile drains a major path for phosphorus loss, studies find | Soil Science Society of America
Why do persistent organic pollutants matter? | UNEP - UN Environment Programme
The 12 Initial POPs
https://www.epa.gov/sites/default/files/2016-09/documents/toxaphene.pdf
Volatile Organic Compounds | American Lung Association
New mercury threat to oceans from climate change - BBC News
Phosphorus - Element information, properties and uses | Periodic Table
CDC | Case Definition: Phosphorus
Phosphorus and Water | U.S. Geological Survey
Overview of Total Maximum Daily Loads (TMDLs) | US EPA
Aldrin/Dieldrin | ToxFAQs™ | ATSDR
Polychlorinated Biphenyls (PCBs) | Public Health Statement | ATSDR
Supreme Court narrows scope of Clean Water Act : NPR
https://fsa.prod.usda.gov/sites/default/files/documents/Final-Report_TNC-Mackinaw_AG-3151-P-12-0113.pdf
Apply for Funding - Grant Opportunities
Search Results Detail | Grants.gov